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Safeband_X
Proposal Document

Group WD025

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1. Background

Personal safety in urban and semi-urban environments has become a growing concern in Sri Lanka, with the growing rate of crimes particularly among university students, women, and individuals who commute or travel alone. While smartphones offer basic safety features such as emergency calling, they are slow to access in a moment of genuine threat — requiring the user to unlock the device, navigate to an app, and initiate contact. This delay can be critical.

The Internet of Things (IoT) has opened new possibilities for building compact, affordable, embedded safety devices that can respond in real time. An IoT-based wearable that automatically captures audio-visual evidence, broadcasts a live video stream, sends GPS-tagged alerts to emergency contacts within seconds, and detects sudden movements or unusual activity through motion sensing represents a significant advancement over existing solutions.

This project proposes Safeband_X, a clip-on IoT safety device built around the ESP32-CAM microcontroller module, augmented with an INMP441 I2S MEMS microphone for synchronized audio recording and an MPU6050 motion sensor for detecting motion and orientation changes. The device is designed to be affordable, portable, and deployable as a prototype consumer product for the Sri Lankan market.

2. Problem and Motivation

2.1 Problem Statement

In today's rapidly changing socio-economic setting, people experience uncertainty concerning time frames and possible dangers. The number of criminal acts, especially crimes against women, has increased, making particular categories of people, susceptible to true safety threats. During emergencies where the individual experiences violence, harassment, or abuse, the victim does not always get enough time to use their mobile phone, unlock it, and call for help. There are no wearable IoT devices that can meet all of the following requirements at the same time:

- Capture audio-visual evidence of the incident
- Automatically notify emergency contacts with live GPS location
- Stream live video to trusted contacts in real time
- Trigger automatically without requiring manual button input, using impact detection via MPU6050
- Trigger a physical deterrent alarm to attract public attention

Current technologies, such as alarms, function only as acoustic signals without any form of evidence collection or contact notification. Smartphone apps are only useful when a person actively uses them in case of danger.

2.2 Motivation

By addressing this problem, Safeband_X offers direct and measurable societal benefits:

- Automatic protection - the MPU6050 detects sudden high-G impacts or violent movement and triggers the SOS system without any user action, covering scenarios where manual activation is impossible
- Dual trigger redundancy - either the panic button or the impact sensor can independently activate the system, ensuring no scenario is left unprotected
- Immediate evidence capture - synchronized audio and video recordings from the moment of the incident provide admissible evidence for law enforcement
- Faster emergency response - GPS-tagged push notifications reach contacts in under 5 seconds, dramatically reducing response time
- Deterrence - a loud siren and visible red LED strobe can deter a potential attacker before harm occurs
- Affordability - the total component cost of approximately LKR 6,000-9,500 makes this accessible to students and everyday commuters

3. Aim and Objectives

3.1 Aim

To design and build a wearable IoT personal safety device that enables users to instantly capture audio-visual evidence, notify emergency contacts with live GPS location, and deter threats triggered either by a manual panic button press or automatically by the MPU6050 impact detection sensor.

3.2 Objectives

The following measurable objectives will be completed to achieve the aim:

1. Implement a 3-second panic button hold detection on the ESP32-CAM
2. Integrate the MPU6050 accelerometer/gyroscope via I2C and implement high-G threshold detection to automatically trigger SOS when a sudden impact or violent movement is detected (threshold >2.5G acceleration)
3. Implement dual-trigger logic - either the panic button or MPU6050 independently activates the full SOS sequence
4. Interface the INMP441 I2S MEMS microphone with the ESP32-CAM for real-time audio capture at 16kHz, 16-bit PCM
5. Implement FreeRTOS dual-core task scheduling - Core 0 for video capture and WiFi, Core 1 for audio sampling and SD card writes
6. Audio and video streams written to the microSD card
7. Capture a JPEG snapshot within 500ms of trigger and upload it to the Blynk app dashboard
8. Host an MJPEG live video stream from the ESP32-CAM accessible via a URL sent in the push notification
9. Integrate the GPS Neo-6M module to obtain real-time coordinates and include a Google Maps link in the Blynk push notification
10. Trigger a piezo siren and RGB LED simultaneously upon activation
11. Implement a safe-confirmation button to cancel false alerts within 10 seconds
12. Log all alert events - trigger source (button or impact), timestamp, GPS coordinates, and SD card filename to EEPROM for post-incident retrieval
13. Power the device from a 3.7V LiPo battery with TP4056 USB charging circuit, targeting a minimum 6-hour operational life

4. System Diagram

Figure 1 below illustrates the overall system architecture of the Safeband_X device, using flowcharts.

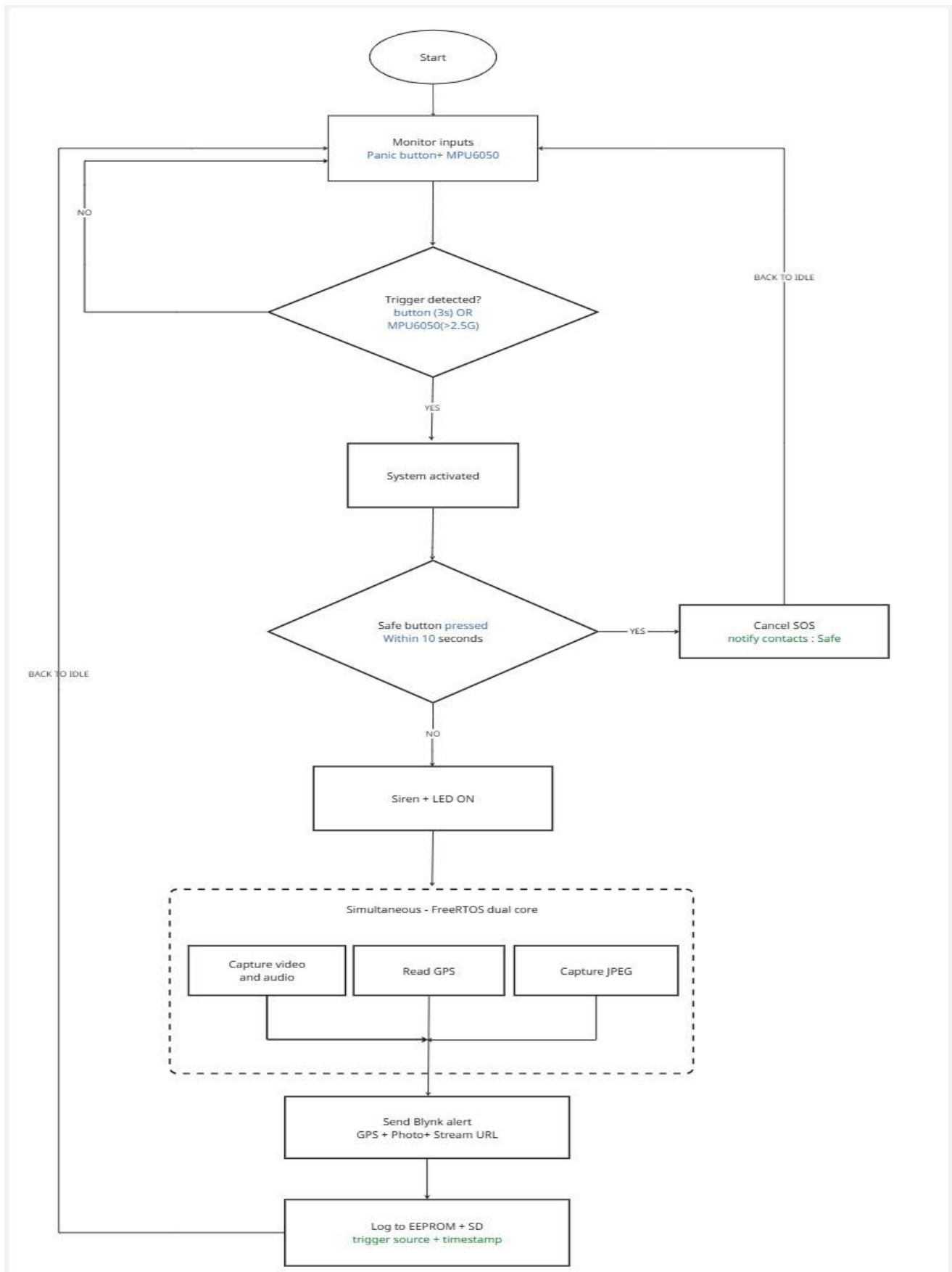


Figure 1: Safeband_X system architecture

The system has two independent trigger inputs: the manual panic button (3-second hold) and the MPU6050 impact sensor (automatic on high-G detection). Either input independently activates the full SOS sequence — piezo siren, RGB LED strobe, AV recording, GPS alert via Blynk, and MJPEG live stream. The ESP32-CAM serves as the central controller running FreeRTOS dual-core scheduling. All events are logged with their trigger source to EEPROM and the microSD card. The device is powered by a rechargeable LiPo battery with TP4056 charging.

5. Methodology

5.1 Hardware

Component	Role	Specification
ESP32-CAM (OV2640)	Main controller + camera + WiFi + SD slot	240MHz dual-core, 4MB PSRAM, 802.11 b/g/n
MPU6050 IMU	Automatic impact / attack detection via I2C	3-axis accel + gyro, I2C address 0x68, interrupt on threshold
INMP441 MEMS mic	Audio capture via I2S protocol	16-bit PCM, 16kHz sample rate, I2S digital output
GPS Neo-6M	Live location via UART	2.5m CEP accuracy, 9600 baud NMEA output
MicroSD card (32GB)	AVI recording + event log storage	Class 10, FAT32 formatted
Piezo siren (active)	Audible deterrent alarm	85dB minimum output
RGB LED + resistors	Visual SOS flash indicator	Red channel, 470-ohm resistor
Tactile buttons x2	Panic trigger + safe cancel	Normally open, debounced in firmware
LiPo 3.7V 1000mAh	Primary power supply	3.7V nominal, 1000mAh capacity
TP4056 + MB board	USB charging + programming	1A charge current, over-discharge protection

Table 1: Hardware components

5.2 Software and Tools

- Arduino IDE 2.x — primary development environment for ESP32-CAM firmware
- ESP32 Arduino Core — hardware abstraction layer and peripheral drivers
- MPU6050 library (by Electronic Cats) — I2C communication and interrupt configuration for impact detection
- TinyGPS++ library — NMEA sentence parsing for GPS coordinate extraction
- Blynk IoT library — push notifications, image widget upload, and mobile dashboard
- ESP32 I2S driver — audio sampling from INMP441 at 16kHz
- ESP32 EEPROM library — persistent event logging across power cycles
- FreeRTOS — dual-core task scheduling for synchronised AV capture
- AVI muxer library — interleaving video frames and PCM audio chunks into AVI container

5.3 Dual Trigger Logic — MPU6050 and Panic Button

An I2C connection (SDA/SCL) links the MPU6050, which has been programmed to issue a hardware interrupt on the INT pin whenever there is an exceedance of a specified acceleration threshold in any direction. For instance, a threshold value of 2.5G is applied. It will be too high for normal walking, but will sense any sudden acceleration caused by pushing, falling, or an impact. The interrupt pin is routed to a GPIO of the ESP32-CAM and is handled by firmware in the same way as the panic button input. As a result, protection can be provided regardless of whether there is no way to interact physically with the device.

In order to lower the energy requirements of the system, it works in the low-power mode in which only the MPU6050 and the panic button operate. The ESP32-CAM itself sleeps while the MPU6050 or pressing the button wakes it up. This ensures continuous monitoring while significantly reducing battery usage.

In the EEPROM event log, we save both a source of triggering ('BUTTON' or 'IMPACT') and timestamps with GPS coordinates of the events.

5.4 Key Technical Approach — Dual-Core AV

The ESP32-CAM's dual Xtensa LX6 cores are leveraged using FreeRTOS task pinning to enable parallel processing. Core 0 handles camera frame capture and WiFi communication, while Core 1 handles I2S audio sampling and SD card writes. Both cores write data to the microSD card using FreeRTOS queues and semaphores to maintain proper temporal ordering of audio and video data during recording.

5.5 IoT Platform

Blynk IoT is used as the cloud platform. Upon trigger, the ESP32-CAM uploads a JPEG snapshot to a Blynk image gallery widget and fires a push notification containing the device name, trigger source, timestamp, GPS coordinates, and Google Maps URL. The MJPEG stream URL is appended so contacts can access live video instantly.

6. Evaluation Method

The system will be evaluated against the following measurable criteria:

Test	Method	Pass Criteria
Panic button detection	Press and hold 10 times for 3 seconds; 10 times for less than 3 seconds	10/10 correct triggers, 0 false positives
MPU6050 impact detection	Apply sudden sharp tap 10 times; walk normally for 2 minutes	10/10 taps trigger SOS; 0 false triggers during walking
MPU6050 threshold tuning	Test at 1.5G, 2.0G, 2.5G, 3.0G thresholds with controlled impacts	Identify optimal threshold with 0 false triggers during normal use
Dual trigger independence	Trigger via button only, then via impact only, then both simultaneously	All 3 scenarios activate full SOS sequence correctly
AV recording quality	Playback recorded AVI files	Audio and video, intelligible audio, no corruption
GPS accuracy	Compare reported coordinates to known location	Location error under 15 metres
Notification speed	Measure button/impact to phone notification (10 trials each)	Average under 5 seconds on WiFi
MJPEG stream	Open stream URL	Stable stream
False alarm cancel	Activate panic, press safe button within 10 seconds	Siren stops, contacts receive cancel notification
EEPROM trigger source log	Trigger 10 via button, 10 via impact; retrieve log	All 20 entries show correct trigger source, timestamp, coordinates
Battery life	Run full GPS + WiFi + MPU6050 polling loop; measure discharge time	Minimum 6 hours continuous operation

Table 2: Evaluation criteria and pass conditions

7. References

- [1] Espressif Systems. (2023). ESP32-CAM product datasheet.
https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

- [2] InvenSense. (2016). INMP441 omnidirectional microphone datasheet.
<https://invensense.tdk.com/wp-content/uploads/2015/02/INMP441.pdf>

- [3] InvenSense. (2013). MPU-6050 product specification revision 3.4. <https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf>

- [4] u-blox. (2011). NEO-6 GPS module datasheet. https://www.u-blox.com/sites/default/files/products/documents/NEO-6_DataSheet_%28GPS.G6-HW-09005%29.pdf

- [5] Blynk Inc. (2024). Blynk IoT platform documentation. <https://docs.blynk.io>

- [6] Espressif Systems. (2024). FreeRTOS on ESP32 — Arduino core documentation.
<https://docs.espressif.com/projects/arduino-esp32/en/latest/>

- [7] Munjal, S., & Sharma, N. (2021). IoT-based women safety device using Arduino. International Journal of Engineering Research & Technology, 10(3), 245-249.